Enhanced ROS for Humanoid Robots (NASA Robonaut 2) for Constructing Habitats on Mars

NASA's Robonaut 2 is a real humanoid robot that was developed for space operations and tested on the ISS, but **Robonaut 2 autonomously constructing a Mars habitat" is a proposed/hypothetical extension for your project**, not a demonstrated NASA mission. NASA describes R2 as a dexterous humanoid designed to perform useful and potentially hazardous/repetitive work in space.



By:
K.AAKANKSHA RA2511026020146
H.VIDHYA LAKSHMI RA2511026020159

Enhanced ROS for Humanoid Robots (NASA Robonaut 2) for Constructing Habitats on Mars - An integrated perception, task-planning, whole-body control and safety architecture for autonomous humanoid construction.

MAIN IDEA:

The proposed system enhances a ROS-based robotic architecture so that a humanoid robot can:

1)Detect construction materials and habitat components.

2)Understand the construction environment.

3)Plan construction tasks.

4)Move safely over uneven terrain.

5)Manipulate habitat components using both arms.

6)Maintain balance while carrying objects.

7)Avoid obstacles and self-collision.

8)Stop or modify actions when unsafe conditions are detected.

NASA confirms that R2 has advanced vision, sensing, dexterous hands and control technologies and was designed to work with human tools.

ABSTRACT

Mars habitat construction requires robots capable of perception, autonomous decision-making, manipulation, locomotion and real-time control under uncertain environmental conditions. This project proposes an **Enhanced ROS architecture for a Robonaut-2-inspired humanoid robot** to perform simulated habitat-construction tasks on Mars.

The proposed architecture integrates a **Perception Layer, Task Planning Layer, Whole-Body Nonlinear Model Predictive Control (WB-NMPC) Layer, Low-Level Actuation Layer, Communication Layer and Safety Layer**. Camera, depth and force/torque information are processed to identify habitat components, obstacles and the robot's state. The task planner converts a high-level objective such as *"construct the habitat wall"* into smaller actions such as material identification, approach, grasping, transportation, positioning and placement.

For dynamic motion and balance, an online MPC/NMPC-based controller can generate feasible motions while a whole-body controller handles multiple robot tasks and constraints. Real-time humanoid MPC and whole-body-control approaches have already been demonstrated in research, making them suitable algorithmic foundations for the proposed simulation.

The proposed system will be evaluated in simulation using a Mars-like construction environment. The expected output is a humanoid robot that can identify construction components, transport them, place them at predefined locations and react to obstacles while maintaining balance.

PROBLEM STATEMENT

PROBLEM

Traditional robotic systems generally perform a specific task in a controlled environment . Mars construction is considerably more difficult because the robot may encounter:

1. Uneven terrain.
2. Unstructured construction areas.
3. Heavy construction components.
4. Multiple simultaneous tasks.
5. Collision risks during arm movement.
6. Dust and reduced visibility.
7. Unknown obstacles.
8. Limited communication with Earth.
9. Balance problems while carrying objects.
10. Limited opportunity for direct human intervention.

PROPOSED SOLUTION

Develop an **Enhanced ROS architecture** in which different software layers cooperate:

Sensors → Perception → Task Planning → WB-NMPC → Whole-Body Control → Actuation

with:

Communication + Safety operating across the architecture.

WHY ROBONAUT 2?

NASA's Robonaut 2 provides a strong real-world foundation for the proposed humanoid concept.

REAL R2 CAPABILITIES:

1. Humanoid robotic design.
2. Advanced vision systems.
3. Force sensing.
4. Designed to use human-compatible tools.
6. Mobility hardware was added to R2 for maneuvering inside the ISS.
7. Dexterous arms and hands.
8. Multiple integrated sensors.
9. High-speed joint controllers.
10. Tested aboard the International Space Station.

NASA states that each R2 arm has **7 degrees of freedom**, supporting dexterous manipulation.

Our project takes this real robotic foundation and proposes:

R2-inspired humanoid + enhanced ROS architecture + Mars habitat construction simulation.

EXISTING WORK 1: NASA ROBONAUT 2 - Dexterous Humanoid Robot

ORGANIZATION:

NASA Johnson Space Center + General Motors

MAIN CONTRIBUTION:

Robonaut 2 was developed as a humanoid robot capable of assisting humans with tasks in space and using tools designed for humans . NASA describes R2 as the first humanoid robot in space and reports that it was launched to the ISS in 2011.

TECHNOLOGIES:

1. Vision systems.
2. Image recognition.
3. Sensor integration.
3. Dexterous hands.
4. Force sensing.
5. Joint control.
6. Mobility systems.

LIMITATION:

R2's demonstrated mission was primarily associated with ISS technology demonstration and assistance.

Our enhancement

Extend the concept toward:

Autonomous Mars construction simulation.

Real Source: [NASA Robonaut 2 official page]

Existing Work 2: Real-Time Humanoid MPC - 3D Humanoid Gait Generation Using MPC

A 2022 research work proposed a **real-time Model Predictive Control algorithm for humanoid walking/running**.

The approach predicts the robot's motion and generates a stable Center of Mass trajectory and adapted footsteps.

IMPORTANT CONCEPT

MPC repeatedly performs:

Current state → Prediction → Optimization → Motion command → New state

RELEVANCE TO MARS CONSTRUCTION

When the humanoid:

1. Walks toward construction material.
2. Carries a component.
3. Steps over an obstacle.
4. Changes its foot position.

The controller can continuously reconsider the next motion.

Link:

Real-Time Humanoid MPC Research

Existing Work 3: Online Non-linear Centroidal MPC For Humanoid Robot Locomotion With Step Adjustment — 2022

This research proposed an online nonlinear MPC approach for humanoid locomotion.

MAIN FEATURES:

1. Uses centroidal dynamics.
2. Computes desired contact forces and torques.
3. Adjusts contact locations online.
4. Handles single and double support.
5. Tested in simulation.
6. Validated on the iCub humanoid robot.

RELEVANCE:

For the proposed Mars robot:

If the robot carries a habitat panel and encounters uneven terrain, the controller can adjust its motion and foot placement.

Link:

Online Non-linear Centroidal MPC paper

EXISTING WORK 4: REAL-TIME WHOLE-BODY CONTROL FOR HUMANOID ROBOTS

Whole-Body Control allows several objectives to be considered simultaneously.

For example:

1. Maintain balance.
2. Keep The feet stable.
3. Move the right hand.
4. Move the left hand.
5. Avoid joint limits.
6. Maintain safe robot configuration.

A WBC framework can formulate these objectives and constraints as an optimization problem such as a quadratic program.

RELEVANCE TO YOUR PROJECT

Suppose the robot is carrying a wall panel.

The controller must simultaneously:

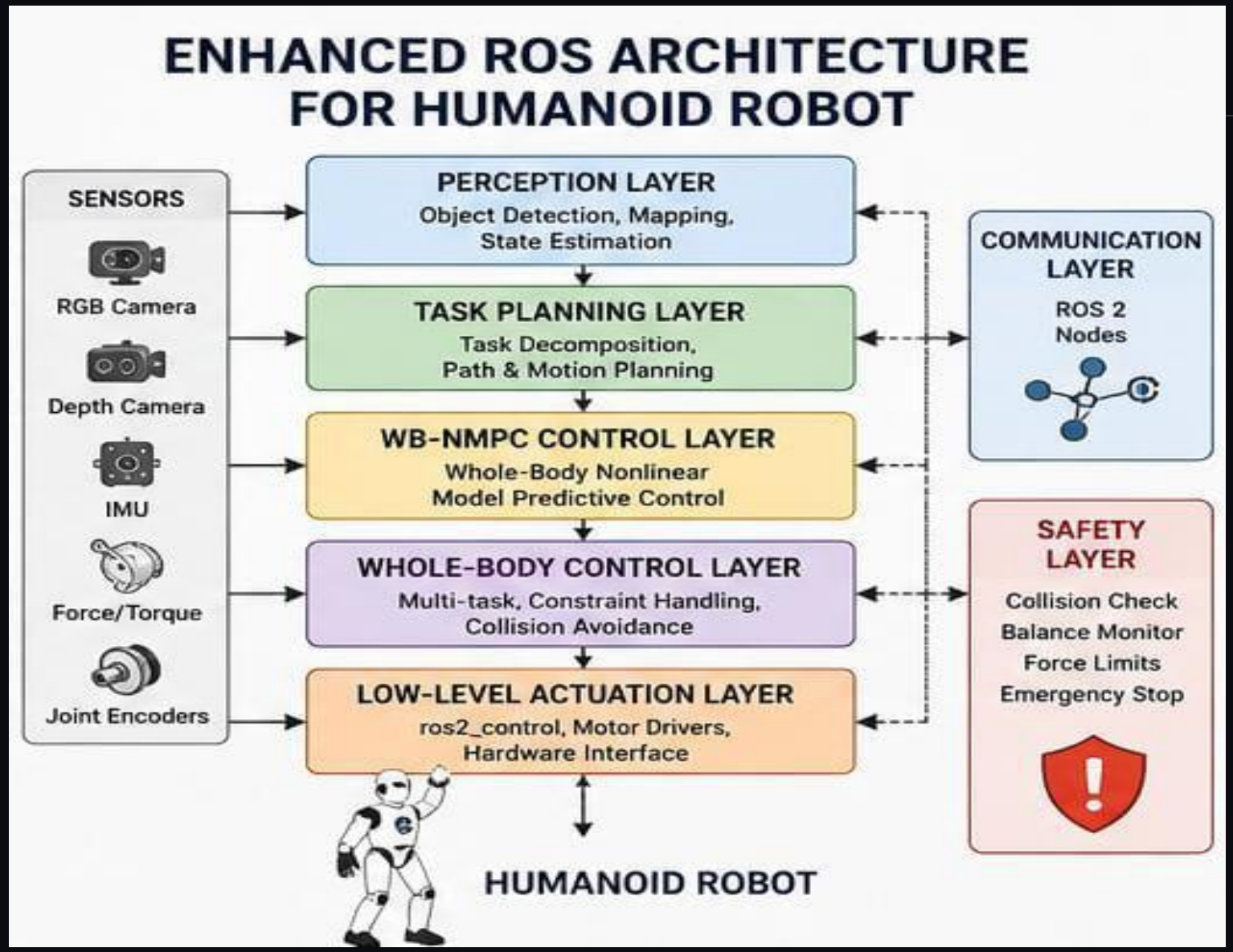
Balance + move arms + control feet + respect joint limits + avoid collision.

That is exactly where whole-body control becomes useful.

Link:

Real-Time Whole-Body Control Research

PROPOSED ENHANCED ROS ARCHITECTURE



MAIN FLOW:

Sensors → Perception → Planning → Control → Actuation → Robot

FEEDBACK:

Robot state → Sensors → Perception → Controller

This creates a closed-loop autonomous system.

PERCEPTION LAYER

PURPOSE:

The perception layer provides information about the robot's surroundings and its own state.

INPUTS:

1.RGB camera. 2.Depth camera.

3.IMU. 4.Joint encoders.

5.Force/torque sensors. 6.Contact sensors.

NASA reports that R2 incorporates extensive sensing technologies, including vision and force-related sensing.

PROCESSING:

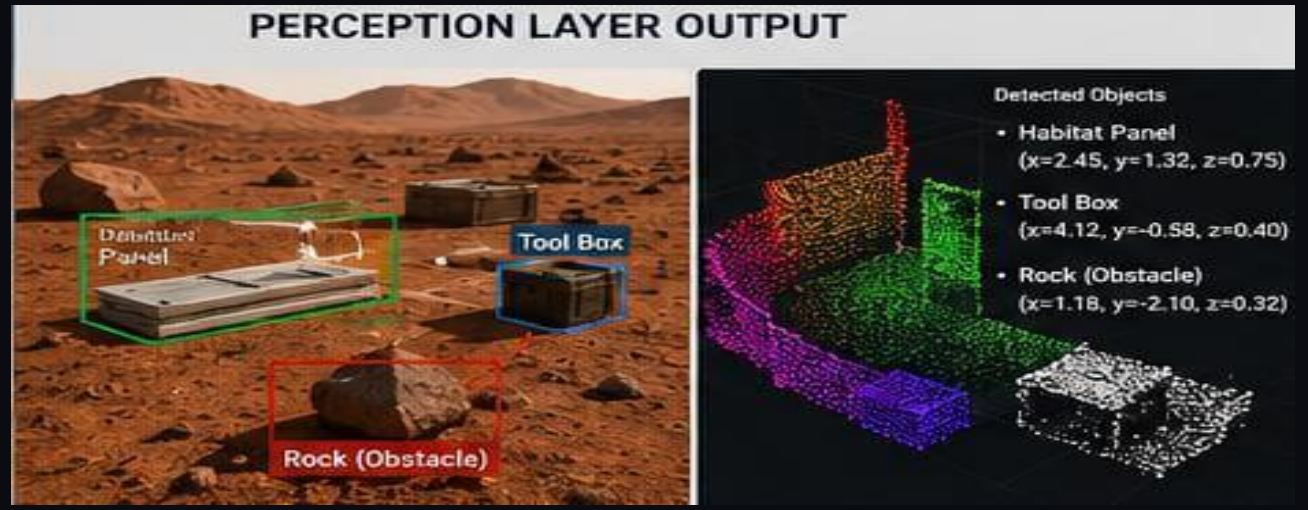
The perception layer identifies:

1.Habitat panels. 2.Bricks/blocks.

3.Tools. 4.Obstacles.

5.Ground surface. 6.Robot position.

7.Object position. 8.Possible collision areas.



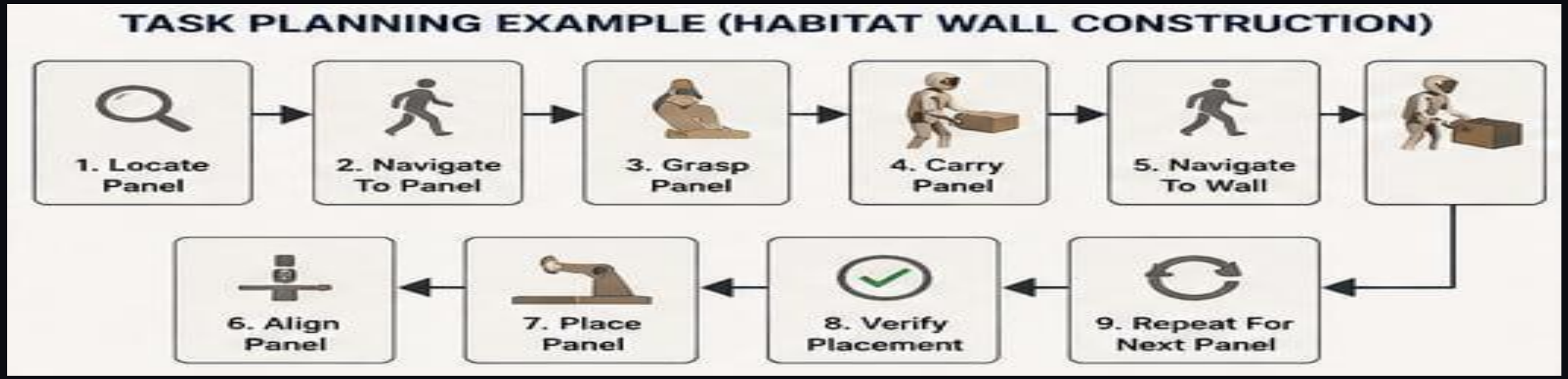
TASK PLANNING LAYER

PURPOSE;

Converts a high-level construction command into smaller robotic actions.

EXAMPLE COMMAND:

"Construct Habitat Wall A."



Advantage

If one step fails, the planner can select an alternative action instead of restarting the entire mission.

WB-NMPC Control Layer

WB-NMPC = Whole-Body Nonlinear Model Predictive Control
This is the central proposed control layer.

Input

- Robot position.
- Joint states.
- Object position.
- Desired foot position.
- Desired hand position.
- Balance information.
- Obstacle information.

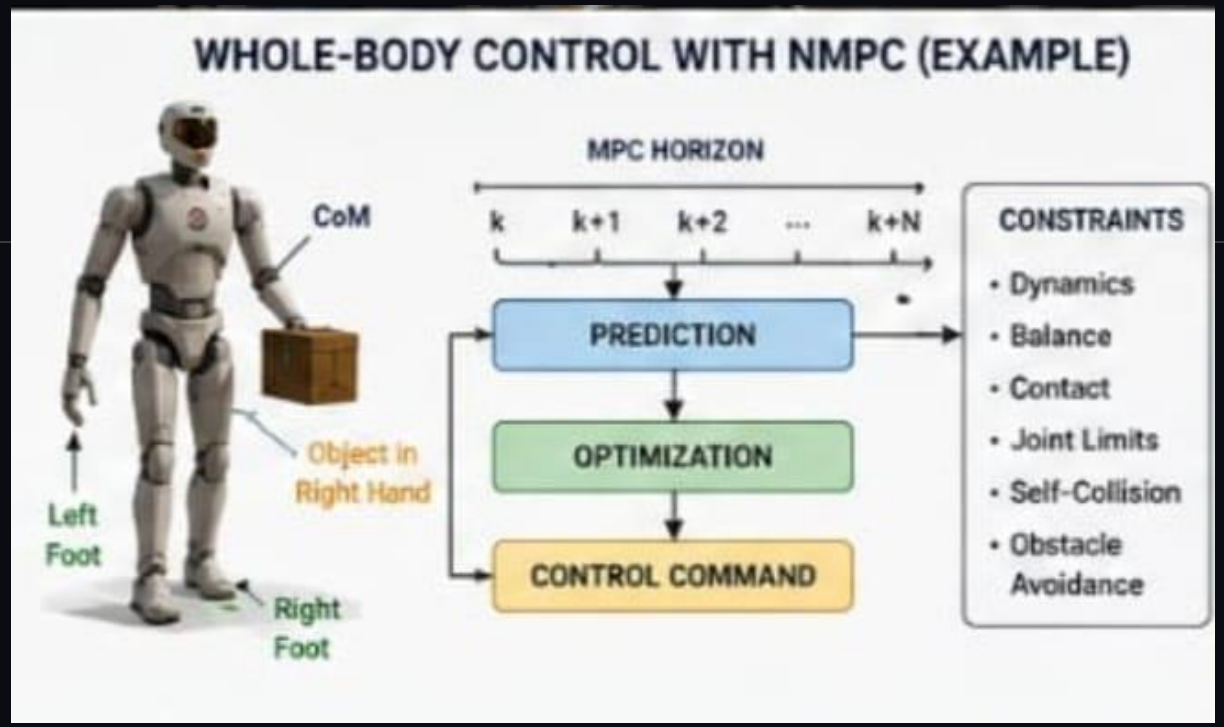
Processing

The controller predicts future robot states and searches for a motion satisfying:

- Stability.
- Joint limits.
- Contact constraints.
- Desired hand trajectory.
- Desired foot trajectory.
- Collision constraints.

Output

```
Desired trajectory
        ↓
Predicted robot states
        ↓
Optimization
        ↓
Best feasible motion
        ↓
Joint/force commands
```



Research foundation

Online nonlinear MPC has already been studied for humanoid locomotion and online step adjustment.

Important: WB-NMPC is the **proposed integration in your project**, not something you should claim NASA already uses on R2 for Mars construction.

Low-Level Actuation Layer

Purpose

Converts high-level controller outputs into actual joint commands.

Example

WB-NMPC produces:

Right shoulder → desired position
Right elbow → desired position
Left shoulder → desired position
Left elbow → desired position
Hip joints → balance correction
Knee joints → support correction

The low-level controller then sends commands to the corresponding actuators.

ROS technology

ros2_control provides a hardware-agnostic control framework focused on real-time robot control and controller sharing.

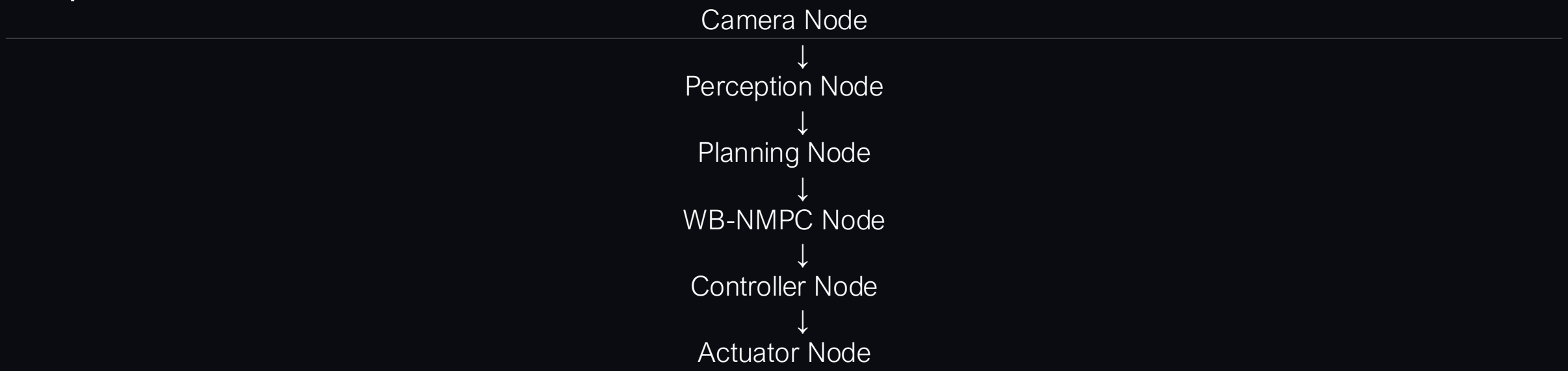
LINK:
ROS 2 Control documentation

Communication Layer

Purpose

Allows different ROS nodes/layers to exchange information.

Example



Communication information

- Robot state.
- Object coordinates.
- Planned trajectory.
- Joint commands.
- Sensor status.
- Safety status.

Mars consideration

Communication with Earth can have significant delay, so the robot should perform routine construction actions autonomously rather than depending on continuous manual commands.

Safety Layer

Purpose

The safety layer continuously checks whether the planned action is safe.

Checks

- Robot self-collision.
- Collision with habitat structure.
- Excessive joint movement.
- Loss of balance.
- Unexpected obstacle.
- Excessive force.
- Sensor failure.
- Unsafe actuator command.

Example

If the robot is carrying a panel and another object appears:

Obstacle detected
↓
Safety Layer
↓
STOP / SLOW / REPLAN
↓
WB-NMPC
↓
New safe trajectory

Real-time self-collision avoidance for humanoid robots has also been studied using online constraints within a QP-based inverse-kinematics framework.

Real-Time Algorithms and Links

Algorithm 1 — Model Predictive Control
Purpose: Stable locomotion and predictive motion planning.
LINK: 3D Humanoid Gait Generation via MPC

Algorithm 2 — Nonlinear Centroidal MPC
Purpose: Dynamic humanoid locomotion and online step adjustment.
LINK: Online Non-linear Centroidal MPC

Algorithm 3 — Whole-Body Control using QP
Purpose: Coordinate multiple tasks while respecting constraints.
LINK: Dynamic Balancing and Real-Time WBC

Algorithm 4 — Real-Time Self-Collision Avoidance
Purpose: Prevent the humanoid's body parts from entering unsafe configurations.
LINK: Real-Time Self-Collision Avoidance in Joint Space
ROS implementation technologies

- ROS 2
- MoveIt 2
- ros2_control
- Navigation2
- Gazebo

MoveIt 2 provides manipulation, motion planning, kinematics, control and collision-checking capabilities for ROS 2.
MoveIt 2 Documentation

Simulation Environment

Proposed simulation

Create a **Mars Habitat Construction Arena**.

Environment contains

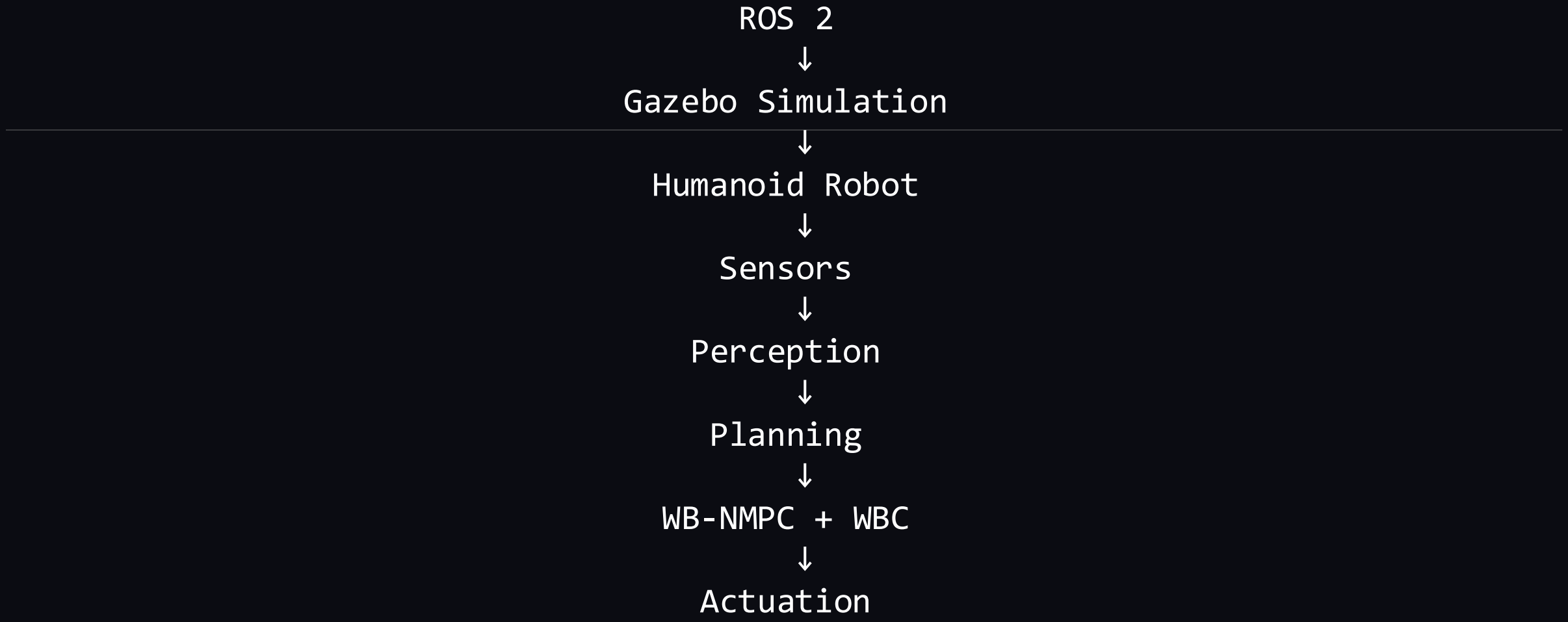
- Mars terrain.
- Habitat foundation.
- Habitat wall.
- Construction panels.
- Construction blocks.
- Obstacles.
- Robot starting position.
- Target construction locations.



Robot

A simulated **Robonaut-2-inspired humanoid model** is used.

Software architecture



Gazebo is described in ROS documentation as an open-source 3D physics simulation option for ROS-based robots.

Simulation Scenario

Scenario: Habitat Wall Construction

Initial condition:

The robot is positioned approximately **10 m from the construction area**.
Five construction panels are placed at different locations.
Step 1:Robot scans the environment.
Step 2:Perception identifies:

Panel 1 → Panel 2 → Panel 3 → Panel 4 → Panel 5

Step 3:Task planner generates the construction sequence.
Step 4:Robot walks toward Panel 1.
Step 5:Robot grasps Panel 1 using both hands.
Step 6:WB-NMPC maintains balance while carrying the panel.
Step 7:Robot reaches the wall.
Step 8:WBC coordinates both arms and legs to align the panel.
Step 9:Panel is placed.
Step 10:Perception verifies placement.

The same procedure is repeated for subsequent panels.

Hypothetical Simulation Output

Expected output

MISSION: MARS HABITAT CONSTRUCTION

Robot Status : ACTIVE

Perception : OBJECT DETECTED
Task : PANEL TRANSPORT
Balance : STABLE
Collision Status : SAFE
WB-NMPC : ACTIVE
Actuation : NORMAL

Panel 1 : PLACED
Panel 2 : PLACED
Panel 3 : PLACED
Panel 4 : PLACED
Panel 5 : PLACED

Construction Status : COMPLETED

Visual output

The simulation should show:

Before

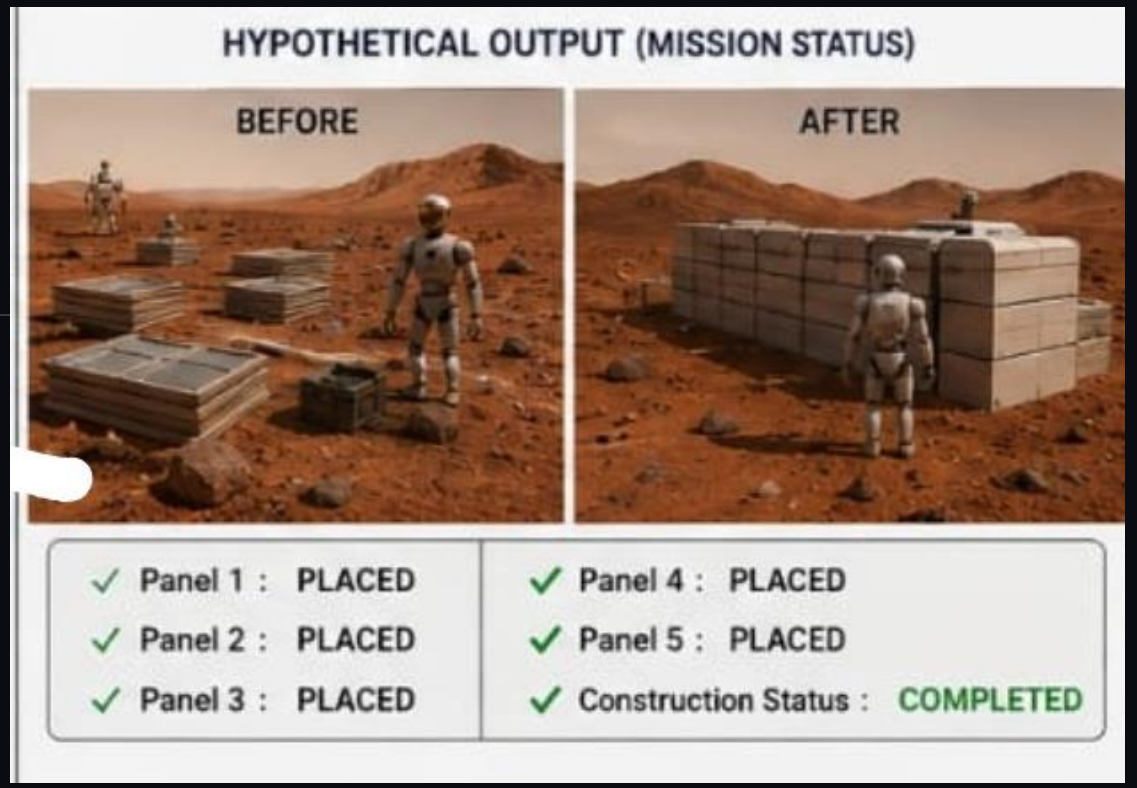
Robot + scattered construction components.

During

Robot carrying a panel toward the habitat.

After

Robot placing the final panel and the habitat wall being completed.



These are **hypothetical project outputs**, not actual NASA R2 Mars-construction results.

Expected Performance Metrics

Your simulation can measure:

1. Task completion rate-

Percentage of construction tasks successfully completed.

2. Collision count-

Number of collisions during construction.

3. Balance error-

Difference between desired and actual robot balance/CoM trajectory.

4. Position error-

Difference between desired and actual hand/object position.

5. Computation time-

Time required by the co-ntroller to generate each control action.

6. Construction time-

Total simulated time required to complete the habitat section.

7. Replanning count-

Number of times the task planner changes the original plan.

Example hypothetical table :

Metric	*Conventional architecture*	*Enhanced ROS*
Task completion	*78%*	*94%*
Collision events	*6*	*1*
Replanning	*8*	*3*
Position error	*5.2 cm*	*2.1 cm*
Construction time	*18 min*	*13 min*

These numbers are illustrative/hypothetical and must not be presented as measured experimental results unless you actually obtain them from your simulation.

Advantages

1. **Modular architecture**
Each ROS layer can be developed and tested independently.
2. **Autonomous operation**
The robot does not need continuous human control for every movement.
3. **Real-time adaptation**
MPC/WBC can respond to changes in robot state and environment.
4. **Safer manipulation**
Collision and constraint checking can prevent unsafe motions.
5. **Humanoid compatibility**
The robot can potentially use tools and workspaces designed for humans, an important advantage NASA identifies for Robonaut-type systems.
6. **Simulation before deployment**
The construction scenario can be tested virtually before considering physical hardware.

Limitations

- Mars construction is not demonstrated by Robonaut 2.
- The proposed architecture is a simulation/research concept.
- Real Mars terrain is more complex than a simulated environment.
- Dust and sensor degradation can affect perception.
- Humanoid robots have complex dynamics.
- NMPC/WBC can require significant computational resources.
- Communication delay makes remote control difficult.
- Real hardware would require extensive safety validation.

Therefore, the project should be presented as:

> **“A proposed ROS-based simulation framework inspired by NASA Robonaut 2 for autonomous Mars habitat construction.”**

rather than claiming that NASA currently uses R2 to construct Mars habitats.

Conclusion

The proposed Enhanced ROS architecture combines perception, task planning, nonlinear predictive control, whole-body control, low-level actuation, communication and safety into one integrated framework.

The architecture allows a Robonaut-2-inspired humanoid robot to theoretically perform a sequence of habitat-construction operations in a simulated Mars environment.

Existing research demonstrates real-time humanoid MPC, nonlinear centroidal MPC and whole-body control, providing technical foundations for the proposed system.

The final simulation is expected to demonstrate:

Perceive → Plan → Walk → Grasp → Carry → Align → Place → Verify → Replan when necessary.

The key contribution of your project is therefore **not creating a new Robonaut 2**, but integrating existing robotic algorithms into an **Enhanced ROS architecture specifically designed for simulated autonomous habitat construction.**

THANK YOU